

**AR-DUINO-M: AUGMENTED REALITY-DRIVEN USER INTERFACE FOR NAVIGATION AND OPERATION OF  
MICROCONTROLLERS**  
**RESEARCH-MODIFIED EVALUATION QUESTIONNAIRE**

*Dear Respondent: Please evaluate the AR-DUINO-M application based on your hands-on experience. Your responses will be treated with strict confidentiality for thesis research purposes. Thank you for your utmost cooperation.*

|                                                                                                                                                              |                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Name (Optional):</b> _____                                                                                                                                | <b>Date:</b> _____                                                                   |
| <b>Respondent:</b> <input type="checkbox"/> Student (End-User) <input type="checkbox"/> Instructor / Expert                                                  | <b>Device:</b> <input type="checkbox"/> Provided <input type="checkbox"/> Own Device |
| <b>Program / Course:</b> _____                                                                                                                               | <b>Year Level:</b> _____                                                             |
| <b>EVALUATION SCALE:</b> [5] Very Much Acceptable (VMA) • [4] Much Acceptable (MA) • [3] Acceptable (A) • [2] Less Acceptable (LA) • [1] Not Acceptable (NA) |                                                                                      |
| <i>Direction: Please place a check mark ( ✓ ) in the box corresponding to your rating for each indicator.</i>                                                |                                                                                      |

| Criteria / Evaluation Indicators                                                                                  | 5 | 4 | 3 | 2 | 1 |
|-------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| <b>A. FUNCTIONALITY</b>                                                                                           |   |   |   |   |   |
| 1. The AR-DUINO-M correctly displays Arduino pin functions and components.                                        |   |   |   |   |   |
| 2. The AR-DUINO-M performs its intended functions accurately.                                                     |   |   |   |   |   |
| 3. The AR-DUINO-M supports interaction with sensors, LEDs, and motors effectively.                                |   |   |   |   |   |
| 4. The AR-DUINO-M provides useful real-time feedback during operation.                                            |   |   |   |   |   |
| <b>B. RELIABILITY</b>                                                                                             |   |   |   |   |   |
| 1. The AR tracking is stable and consistent during use.                                                           |   |   |   |   |   |
| 2. The AR-DUINO-M runs smoothly without lag or crashes.                                                           |   |   |   |   |   |
| 3. The AR-DUINO-M performs well under normal usage conditions.                                                    |   |   |   |   |   |
| 4. The AR-DUINO-M can continuously operate without unexpected errors.                                             |   |   |   |   |   |
| 5. The AR-DUINO-M recovers properly after interruptions or errors.                                                |   |   |   |   |   |
| <b>C. USABILITY</b>                                                                                               |   |   |   |   |   |
| 1. The AR-DUINO-M is easy to learn and use, with clear instructions and prompts for beginners.                    |   |   |   |   |   |
| 2. Navigation and interaction with virtual components and controls are simple and intuitive.                      |   |   |   |   |   |
| 3. The AR-DUINO-M minimizes confusion when performing tasks.                                                      |   |   |   |   |   |
| 4. The design, layout, and interface elements (buttons, menus, labels) are visually appealing and well-organized. |   |   |   |   |   |
| 5. The AR overlays are clear and provide an immersive, engaging AR experience.                                    |   |   |   |   |   |

| Criteria / Evaluation Indicators (Continued)                                                                              | 5 | 4 | 3 | 2 | 1 |
|---------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| <b>D. EFFICIENCY</b>                                                                                                      |   |   |   |   |   |
| 1. The AR-DUINO-M responds quickly to user actions.                                                                       |   |   |   |   |   |
| 2. The AR-DUINO-M loads AR features within an acceptable time.                                                            |   |   |   |   |   |
| 3. Resource usage such as battery and memory consumption is efficient.                                                    |   |   |   |   |   |
| 4. The AR-DUINO-M maintains good performance during prolonged use.                                                        |   |   |   |   |   |
| 5. The AR-DUINO-M efficiently processes user inputs and outputs.                                                          |   |   |   |   |   |
| <b>E. PORTABILITY</b>                                                                                                     |   |   |   |   |   |
| 1. The AR-DUINO-M ran smoothly on the device I used.                                                                      |   |   |   |   |   |
| 2. The AR-DUINO-M's requirements (storage, sensors, camera) are minimal and can generally be met by many Android devices. |   |   |   |   |   |
| 3. The AR-DUINO-M can be installed and used easily on supported devices.                                                  |   |   |   |   |   |
| 4. The AR-DUINO-M worked properly without requiring extra setup or configuration on my device.                            |   |   |   |   |   |
| <b>F. MAINTAINABILITY [ For Instructors / Subject Matter Experts only ]</b>                                               |   |   |   |   |   |
| 1. The AR-DUINO-M design allows easy modification and improvement of features.                                            |   |   |   |   |   |
| 2. Errors and issues in the AR-DUINO-M can be easily identified and corrected.                                            |   |   |   |   |   |
| 3. The AR-DUINO-M structure supports future updates and enhancements.                                                     |   |   |   |   |   |
| 4. The AR-DUINO-M components are organized properly for maintenance purposes.                                             |   |   |   |   |   |
| 5. The AR-DUINO-M can be improved without affecting overall functionality.                                                |   |   |   |   |   |
| <b>G. EDUCATIONAL EFFECTIVENESS [ For Students / End-Users only ]</b>                                                     |   |   |   |   |   |
| 1. The AR-DUINO-M improves my understanding of Arduino and microcontrollers.                                              |   |   |   |   |   |
| 2. AR visualization helps me connect theory with actual hardware.                                                         |   |   |   |   |   |
| 3. The AR-DUINO-M application enhances my problem-solving skills.                                                         |   |   |   |   |   |
| 4. The AR-DUINO-M increases my engagement and interest in learning.                                                       |   |   |   |   |   |
| 5. The AR-DUINO-M is an effective tool for hands-on learning.                                                             |   |   |   |   |   |

**Comments / Suggestions / Recommendations for Improvement:**

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*Thank you for your valuable time and participation in our thesis research!*